

Microeconomic Analysis

Homework 1 – Solutions

Instructor: Mark Dean

Solutions prepared by Hao Jiang

Due Thursday 17th September

Definitions used in the solutions. The homework refers to the definitions of α , β , and WARP from class. We use them in the following form. A choice correspondence is nonempty-valued and satisfies $C(A) \subseteq A$ for every nonempty menu A .

$$\alpha : \quad x \in C(A), \quad x \in B \subseteq A \quad \implies \quad x \in C(B).$$

Thus a chosen alternative must remain chosen after a contraction of the menu, provided that the alternative itself remains feasible.

$$\beta : \quad x, y \in C(A), \quad A \subseteq B, \quad y \in C(B) \quad \implies \quad x \in C(B).$$

Thus, if x and y were both chosen from a smaller menu, then an expansion cannot keep y chosen while dropping x .

We use the class formulation of WARP:

$$x \in C(A), \quad y \in C(B), \quad x, y \in A \cap B \quad \implies \quad x \in C(B).$$

Because the names of A, B and x, y can be interchanged, the same condition also implies $y \in C(A)$ under the same premises.

Question 1 Some things to clear up from class

1. Come up with a choice correspondence that satisfies property α but not property β , and another that satisfies β but not α .

Solution. We need two separate examples. In each example we must define the choice correspondence on *every* nonempty subset of the underlying set, and then verify one axiom while displaying a concrete violation of the other.

(a) **A correspondence satisfying α but not β .** Let

$$X = \{x, y, z\}.$$

Define C_α by

A	$C_\alpha(A)$
$\{x, y, z\}$	$\{x\}$
$\{x, y\}$	$\{x, y\}$
$\{x, z\}$	$\{x\}$
$\{y, z\}$	$\{y\}$
$\{x\}$	$\{x\}$
$\{y\}$	$\{y\}$
$\{z\}$	$\{z\}$.

We first verify α . The axiom says that whenever an alternative is chosen from a menu, it must remain chosen from every smaller menu that still contains it. Since X has only three alternatives, we can check all nontrivial possibilities.

From the largest menu $\{x, y, z\}$, the only chosen alternative is x . The smaller menus containing x are

$$\{x, y\}, \quad \{x, z\}, \quad \{x\}.$$

By construction,

$$x \in C_\alpha(\{x, y\}), \quad x \in C_\alpha(\{x, z\}), \quad x \in C_\alpha(\{x\}).$$

So every contraction from the largest menu satisfies α .

From $\{x, y\}$, both x and y are chosen. The only strict contractions that retain the relevant chosen alternative are their singleton menus, and

$$C_\alpha(\{x\}) = \{x\}, \quad C_\alpha(\{y\}) = \{y\}.$$

From $\{x, z\}$ only x is chosen and x remains chosen from $\{x\}$. From $\{y, z\}$ only y is chosen and y remains chosen from $\{y\}$. Singleton menus have no nontrivial contractions. Therefore C_α satisfies α .

Now check β . Take

$$A = \{x, y\}, \quad B = \{x, y, z\}.$$

Then $A \subseteq B$ and

$$x, y \in C_\alpha(A) = \{x, y\}.$$

In the larger menu,

$$C_\alpha(B) = \{x\}.$$

Thus x remains chosen but y does not. Apply the definition of β using the pair of alternatives (y, x) : both y and x are in $C_\alpha(A)$ and $x \in C_\alpha(B)$, so β would require $y \in C_\alpha(B)$. This is false. Hence β fails.

(b) A correspondence satisfying β but not α . Again let $X = \{x, y, z\}$, but now define C_β by

A	$C_\beta(A)$
$\{x, y, z\}$	$\{x, y\}$
$\{x, y\}$	$\{x\}$
$\{x, z\}$	$\{x\}$
$\{y, z\}$	$\{y\}$
$\{x\}$	$\{x\}$
$\{y\}$	$\{y\}$
$\{z\}$	$\{z\}$.

Property α fails immediately. In the large menu,

$$y \in C_\beta(\{x, y, z\}).$$

The smaller menu $\{x, y\}$ still contains y , but

$$y \notin C_\beta(\{x, y\}) = \{x\}.$$

This is exactly a violation of α .

It remains to verify β . Suppose $A \subseteq B$ and suppose

$$r, s \in C_\beta(A), \quad s \in C_\beta(B).$$

If A is a singleton or a two-element menu, then $C_\beta(A)$ is a singleton. Hence $r = s$. The desired conclusion $r \in C_\beta(B)$ is then identical to the premise $s \in C_\beta(B)$.

The only menu at which two distinct alternatives are chosen is $A = X$, where $C_\beta(X) = \{x, y\}$. But X has no strict superset in the domain. If $A = B = X$, the conclusion of β is again already contained in the premises. Therefore no violation of β is possible, and C_β satisfies β .

The two examples show that neither α nor β implies the other.

2. Show that if a choice correspondence has a utility representation then it must satisfy α and β .

Solution. Suppose C has a utility representation. By definition, there exists a function

$$u : X \rightarrow \mathbb{R}$$

such that for every nonempty menu A ,

$$C(A) = \arg \max_{z \in A} u(z) = \{z \in A : u(z) \geq u(w) \text{ for every } w \in A\}.$$

We prove the two axioms separately.

Proof of α . Take arbitrary menus $B \subseteq A$ and an alternative x satisfying

$$x \in C(A), \quad x \in B.$$

Because $x \in C(A)$, x maximizes u over A . Therefore

$$u(x) \geq u(z) \quad \text{for every } z \in A.$$

Since $B \subseteq A$, every element of B is also an element of A . Hence the same inequalities hold on the smaller menu:

$$u(x) \geq u(z) \quad \text{for every } z \in B.$$

Also $x \in B$, so x is feasible in the smaller menu. Therefore x maximizes u over B , which means

$$x \in C(B).$$

This is exactly α .

Proof of β . Take arbitrary menus $A \subseteq B$ and alternatives x, y satisfying

$$x, y \in C(A), \quad y \in C(B).$$

Since x and y are both utility maximizers on the same smaller menu A , they must attain the same maximal utility level:

$$u(x) = u(y) = \max_{z \in A} u(z).$$

The additional premise $y \in C(B)$ says that y maximizes utility on the larger menu B . Hence

$$u(y) \geq u(z) \quad \text{for every } z \in B.$$

Using $u(x) = u(y)$, we obtain

$$u(x) \geq u(z) \quad \text{for every } z \in B.$$

Since $x \in A \subseteq B$, x is feasible in B . Thus x also maximizes u on B and

$$x \in C(B).$$

This is exactly β .

Therefore every utility-maximizing choice correspondence satisfies both α and β .

3. Prove that a choice correspondence satisfies properties α and β if and only if it satisfies WARP.

Solution. There are two directions to prove.

Direction 1: α and β imply WARP. Assume that C satisfies both α and β . We must prove WARP.

Take arbitrary menus A, B and arbitrary alternatives x, y such that

$$x \in C(A), \quad y \in C(B), \quad x, y \in A \cap B.$$

The goal is to prove

$$x \in C(B).$$

Because $A \cap B \subseteq A$, because $x \in C(A)$, and because $x \in A \cap B$, property α implies

$$x \in C(A \cap B).$$

Similarly, $A \cap B \subseteq B$, $y \in C(B)$, and $y \in A \cap B$, so another application of α gives

$$y \in C(A \cap B).$$

Thus both x and y are chosen from the smaller common menu $A \cap B$.

Now compare the menus $A \cap B$ and B . We have

$$A \cap B \subseteq B,$$

we have just shown

$$x, y \in C(A \cap B),$$

and the original premise gives

$$y \in C(B).$$

Therefore β , applied with the smaller menu $A \cap B$ and the larger menu B , implies

$$x \in C(B).$$

This is precisely WARP.

Direction 2: WARP implies α . Now assume that C satisfies WARP. Take $x \in C(A)$ and suppose

$$x \in B \subseteq A.$$

We must show $x \in C(B)$.

Because a choice correspondence is nonempty-valued, $C(B) \neq \emptyset$. Choose any

$$y \in C(B).$$

Since $B \subseteq A$, we have $A \cap B = B$. Both x and y belong to B : $x \in B$ by assumption, and $y \in C(B) \subseteq B$. Hence

$$x, y \in A \cap B.$$

We also have $x \in C(A)$ and $y \in C(B)$. WARP therefore implies

$$x \in C(B).$$

This is α .

Direction 3: WARP implies β . Again assume WARP. Take $A \subseteq B$ and suppose

$$x, y \in C(A), \quad y \in C(B).$$

We must prove $x \in C(B)$.

Because $x, y \in C(A) \subseteq A$ and $A \subseteq B$, both alternatives belong to

$$A \cap B = A.$$

We have $x \in C(A)$ and $y \in C(B)$. WARP applied directly to the two menus therefore gives

$$x \in C(B).$$

This is β .

We have proved both implications. Hence a choice correspondence satisfies α and β exactly when it satisfies WARP.

4. Here is an alternative definition of WARP, which we will call WARP*.

Axiom 1 (WARP*). For any set S , there exists a b^* in S such that, for any T including b^* , if $C(T) \in S$ then $C(T) = \{b^*\}$.

Show that, if we restrict attention to choice functions (i.e. C is single valued), a data set satisfies WARP* if and only if it satisfies WARP. Is this still the case if we allow for C to be a choice correspondence?

Solution. We first handle the case explicitly requested in the first sentence: C is a choice function. It is then convenient to write $C(A)$ for the chosen *alternative*, rather than repeatedly writing the singleton $\{C(A)\}$.

WARP implies WARP*. Assume WARP and fix an arbitrary nonempty set S . We need to exhibit an element $b^* \in S$ that has the WARP* property.

A natural candidate is the alternative actually chosen from S :

$$b^* := C(S).$$

Because C is a choice function, $C(S) \in S$, so indeed $b^* \in S$.

Now take an arbitrary menu T such that

$$b^* \in T$$

and suppose the premise of WARP* holds:

$$C(T) \in S.$$

Let

$$y := C(T).$$

Then $y \in S$ by the premise, and $y \in T$ because a choice must be feasible. Also $b^* \in S$ because $b^* = C(S)$, and $b^* \in T$ by assumption. Therefore

$$b^*, y \in S \cap T.$$

Moreover,

$$b^* = C(S), \quad y = C(T).$$

Applying WARP to menus S and T gives

$$b^* \in C(T).$$

Since C is single-valued and $C(T) = y$, this can only happen if

$$y = b^*.$$

Thus

$$C(T) = b^*.$$

Since T was arbitrary, the required WARP* condition is satisfied.

WARP* implies WARP. Assume WARP*. We prove WARP by contradiction.

Suppose WARP fails. Then there are menus A, B and alternatives $x, y \in A \cap B$ such that

$$C(A) = x, \quad C(B) = y, \quad x \neq y.$$

The inequality $x \neq y$ follows because, if $x = y$, the alleged failure of WARP disappears.

Consider the two-element set

$$S = \{x, y\}.$$

WARP* says that there exists some $b^* \in S$ with the stated property. Since S has only two elements, there are only two cases.

If $b^* = x$, use the menu $T = B$. We have $x \in B$ because $x \in A \cap B$, and

$$C(B) = y \in S.$$

WARP* would therefore require

$$C(B) = x,$$

contradicting $C(B) = y$.

If $b^* = y$, use the menu $T = A$. We have $y \in A$ because $y \in A \cap B$, and

$$C(A) = x \in S.$$

WARP* would therefore require

$$C(A) = y,$$

contradicting $C(A) = x$.

Both possible candidates fail, contradicting WARP*. Therefore WARP cannot fail. So WARP* implies WARP.

Consequently, for choice functions, WARP and WARP* are equivalent.

What if C is a choice correspondence?

As written, WARP* uses the condition

$$C(T) \in S.$$

When C is a choice function, $C(T)$ denotes a single alternative, so this is well-defined. If C is instead a choice correspondence, then $C(T)$ is a set of alternatives. A natural extension of the premise is therefore

$$C(T) \subseteq S.$$

Under this correspondence-valued version, WARP* becomes:

For every nonempty S , there exists $b^* \in S$

such that, for every T containing b^* ,

$$C(T) \subseteq S \implies C(T) = \{b^*\}.$$

This condition is no longer equivalent to WARP.

To see this, let

$$X = \{x, y\}$$

and define

$$C(\{x\}) = \{x\}, \quad C(\{y\}) = \{y\}, \quad C(\{x, y\}) = \{x, y\}.$$

This correspondence is generated by the constant utility function

$$u(x) = u(y) = 0.$$

Hence it satisfies α and β , and therefore satisfies WARP.

Now take

$$S = \{x, y\}.$$

There are only two possible candidates for b^* .

If $b^* = x$, take

$$T = S = \{x, y\}.$$

Then T contains x and

$$C(T) = \{x, y\} \subseteq S.$$

WARP* would therefore require

$$C(T) = \{x\},$$

which is false.

If instead $b^* = y$, take the same menu $T = S$. Again

$$C(T) = \{x, y\} \subseteq S,$$

so WARP* would require

$$C(T) = \{y\},$$

which is also false.

Thus no $b^* \in S$ satisfies the correspondence version of WARP*. Therefore WARP does not imply WARP* once multi-valued choice is allowed.

The source of the failure is that WARP permits several alternatives to be chosen simultaneously, for example because they are indifferent, whereas WARP* requires the relevant choice set to collapse to the singleton b^* .

Question 2 Utility maximization is not the only choice procedure that is consistent with α and β . There are also other choice procedures that will satisfy these conditions and so are indistinguishable from rational choice. Consider the following decision making procedures. Prove whether or not they will result in choices that satisfy α and β (feel free to assume that there are a finite number of alternatives in each case). If they are consistent with α and β , and therefore admit a utility representation, discuss the relationship between such representations and their ‘true’ utility (if the decision making model has such a thing).

1. Satisficing: A decision maker (DM) is choosing between books from a set B . They have a utility function $u : B \rightarrow \mathbb{R}$, and a ‘threshold utility’ level u^* . In any choice set, they search through the books alphabetically by title, and choose the first book that has utility level u that is equal to or above u^* . If they have not found any such book by the time they reach the end of the choice set, they will choose the book with the highest utility (to make things simpler you can assume that there is no indifference - i.e. no two books have the same utilities).

Solution. Let

$$G := \{b \in B : u(b) \geq u^*\}$$

be the set of books that meet the threshold. Call these the *acceptable* books. The procedure has two branches:

- (i) if a menu contains at least one acceptable book, choose the alphabetically first acceptable book;
- (ii) if a menu contains no acceptable book, choose the book with highest true utility u .

We check α directly.

Take menus $D \subseteq A$ and suppose

$$x = C(A), \quad x \in D.$$

We must prove $C(D) = x$.

Case 1: A contains at least one acceptable book. Then the rule chooses the alphabetically first acceptable book in A . Therefore

$$x \in G,$$

and there is no acceptable book in A whose title comes alphabetically before the title of x .

Because $D \subseteq A$ and $x \in D$, D also contains the acceptable book x . Hence the procedure uses the acceptable-book branch in D as well. Could there be an acceptable book in D alphabetically before x ? No: any such book would also belong to A , contradicting the fact that x was the first acceptable book in A . Therefore x remains the first acceptable book in D , so

$$C(D) = x.$$

Case 2: A contains no acceptable book. Then

$$A \cap G = \emptyset.$$

Since $D \subseteq A$, it follows that

$$D \cap G = \emptyset.$$

Thus both menus use the fallback rule of maximizing true utility. Since $x = C(A)$,

$$u(x) > u(z) \quad \text{for every } z \in A \setminus \{x\},$$

where strict inequality follows from the no-indifference assumption. Every $z \in D \setminus \{x\}$ is also in A , so the same inequalities hold in D . Hence x is the unique highest-utility book in D and

$$C(D) = x.$$

The two cases exhaust all possibilities, so α holds.

Now consider β . Under the assumptions in the question, the procedure is single-valued: alphabetical order identifies a unique first acceptable book, and if no book reaches the threshold then the no-indifference assumption identifies a unique utility maximizer. For any single-valued choice function, β is automatic. Indeed, if

$$x, y \in C(A),$$

then single-valuedness implies $x = y$. If additionally $y \in C(B)$, then $x = y \in C(B)$, which is the conclusion of β .

It is useful to see why the whole procedure is rationalizable. Define a fixed ranking $\succ^*$ on books as follows:

- every acceptable book is ranked above every unacceptable book;
- among acceptable books, the alphabetically earlier title is ranked higher;
- among unacceptable books, the book with higher true utility u is ranked higher.

This ranking does not depend on the menu. The satisficing procedure always chooses its highest-ranked available book according to $\succ^*$. Because the set of books is finite, any strict ranking can be represented numerically: assign the worst book value 1, the next book value 2, and so on. Call the resulting representation $\tilde{u}$. Then

$$C(A) = \arg \max_{x \in A} \tilde{u}(x)$$

for every menu A .

However, $\tilde{u}$ is generally not the DM's true utility u . Below the threshold, the procedure follows u . Once alternatives meet the threshold, the procedure ignores utility differences and follows alphabetical order. Thus the representing utility $\tilde{u}$ rationalizes the *observed choice procedure*; it need not represent the DM's welfare ranking.

2. Elimination by Aspects: The DM is choosing between kettles. Each kettle is defined by its price, its energy efficiency (on a scale from 1 to 5) and its attractiveness (on a scale of 1 to 3). When given a choice set the DM
 - (a) Selects all the kettles with the highest attractiveness rating;
 - (b) From those kettles, selects those with the best energy efficiency;
 - (c) From those that remain, they choose the cheapest.

Solution. For a kettle x , write

$$a(x) = \text{attractiveness}, \quad e(x) = \text{energy efficiency}, \quad p(x) = \text{price}.$$

The procedure compares kettles lexicographically using the vector

$$(a(x), e(x), -p(x)).$$

That is, attractiveness receives first priority; energy efficiency matters only when attractiveness ties; price matters only when both attractiveness and efficiency tie.

We first prove α directly.

Suppose

$$x \in C(A), \quad x \in D \subseteq A.$$

Because x survives stage 1 in A , it has maximal attractiveness in A :

$$a(x) \geq a(z) \quad \text{for every } z \in A.$$

Therefore the same inequality holds for every $z \in D$. Hence x survives stage 1 in D .

Among the stage-1 survivors in A , x has maximal energy efficiency. Any stage-1 survivor in D is also an element of A with maximal attractiveness: because x itself remains in D , the maximal attractiveness level in D is exactly $a(x)$, the same level as in A . Therefore every stage-1 survivor of D was also a stage-1 survivor of A . Since x had weakly greater efficiency than all of them in A , it also has weakly greater efficiency than all of them in D . Hence x survives stage 2 in D .

Finally, among the stage-2 survivors in A , x is cheapest. Every stage-2 survivor in D is also a stage-2 survivor in A , by the same reasoning. Therefore no stage-2 survivor in D has a lower price than x . Hence x survives stage 3 and belongs to $C(D)$. Thus α holds.

Now prove β . Suppose

$$x, y \in C(A), \quad A \subseteq B, \quad y \in C(B).$$

Because both x and y are final choices from A , they must have exactly the same values on all three criteria used by the procedure. More explicitly:

- both have the maximal attractiveness level in A , so $a(x) = a(y)$;
- both survive the efficiency stage, so $e(x) = e(y)$ at the maximal efficiency level among the most attractive kettles;
- both survive the price stage, so $p(x) = p(y)$ at the minimal price among the first two stages' survivors.

Now $y \in C(B)$. Suppose, for contradiction, that $x \notin C(B)$. Then x must be eliminated at some stage in B .

It cannot be eliminated for attractiveness, because $a(x) = a(y)$ and y survives that stage. It cannot be eliminated for efficiency, because $e(x) = e(y)$ and both have the same attractiveness. It cannot be eliminated for price, because $p(x) = p(y)$ and both have the same values of the first two attributes. Therefore x survives every stage whenever y does. Hence

$$x \in C(B).$$

So β holds.

The procedure is therefore consistent with α and β . It can also be understood as maximization of a fixed lexicographic ordering. Since the set of alternatives is finite, that ordering has a real-valued representation $\tilde{u}$. One can construct it simply by ranking the distinct attribute triples lexicographically and assigning increasing integers to successive ranks.

The question does not specify a separate “true” utility for this DM. If the DM did have a compensatory welfare utility in which, for example, a sufficiently large price saving could compensate for one unit of attractiveness, then the rationalizing $\tilde{u}$ would generally not equal that welfare utility. $\tilde{u}$ encodes the procedural priority “attractiveness first, then efficiency, then price.”

3. The DM ranks the alternatives according to price, then looks at the three cheapest and chooses the one with the highest utility.

Solution. For a menu A , let $S_3(A)$ denote the set of the three cheapest alternatives in A , or all alternatives if $|A| \leq 3$. The rule is

$$C(A) = \arg \max_{x \in S_3(A)} u(x).$$

The key feature is that the shortlist $S_3(A)$ itself depends on the menu. Contracting a menu can allow an alternative that was previously the fourth-cheapest to enter the shortlist.

α **fails.** Take four alternatives a, b, x, d with strictly ordered prices

$$p(a) < p(b) < p(x) < p(d)$$

and utilities satisfying

$$u(d) > u(x) > u(b) > u(a).$$

Consider the four-element menu

$$A = \{a, b, x, d\}.$$

The three cheapest alternatives are

$$S_3(A) = \{a, b, x\}.$$

Among these three, x has the highest utility. Therefore

$$C(A) = x.$$

Now contract the menu to

$$D = \{b, x, d\}.$$

The chosen alternative x remains feasible, so this contraction is relevant for α . Since D contains exactly three alternatives,

$$S_3(D) = D = \{b, x, d\}.$$

Now d enters the shortlist, and by construction

$$u(d) > u(x) > u(b).$$

Hence

$$C(D) = d.$$

Thus

$$x = C(A), \quad x \in D \subseteq A, \quad x \neq C(D),$$

which is a direct violation of α .

β under the literal “chooses the one” interpretation. If the highest-utility member of every shortlist is unique, the procedure is a choice function. Then β holds automatically. To see this from the definition, suppose

$$x, y \in C(A).$$

Single-valuedness forces $x = y$. Therefore, whenever the remaining premise $y \in C(B)$ holds, we automatically have $x \in C(B)$ as well.

So under the natural single-valued interpretation of the wording “chooses the one with the highest utility,” the correct classification is:

$$\alpha \text{ fails,} \quad \beta \text{ holds.}$$

Because α fails, the procedure cannot be represented as maximization of a single utility function that is independent of the menu.

If utility ties are allowed and all tied shortlist maximizers are chosen. Then the rule is a genuine correspondence and β can also fail. For completeness, consider the small menu

$$A = \{a, y, x\}$$

with

$$p(a) < p(y) < p(x), \quad u(x) = u(y) > u(a).$$

All three alternatives are shortlisted, so

$$C(A) = \{x, y\}.$$

Now add a new very cheap alternative w with

$$p(w) < p(a), \quad u(w) < u(y).$$

Let

$$B = \{w, a, y, x\}.$$

The three cheapest alternatives are w, a, y , so x is pushed out of the shortlist. Since y has higher utility than w and a ,

$$C(B) = \{y\}.$$

We therefore have

$$x, y \in C(A), \quad A \subseteq B, \quad y \in C(B), \quad x \notin C(B),$$

which violates β .

The economic point is the same in both interpretations: the DM's true utility is used only *after* a menu-dependent price screen. The observed choice rule therefore need not reveal the ranking generated by the true u over the full menu.

4. The DM has a true utility function for kettles based on price, energy efficiency and attractiveness. However, to save time, they always just buy the cheapest kettle.

Solution. The observed rule ignores the DM's stated true utility and uses only price. If ties at the lowest price are allowed, the natural choice correspondence is

$$C(A) = \arg \min_{x \in A} p(x).$$

Equivalently,

$$C(A) = \arg \max_{x \in A} \{-p(x)\}.$$

Thus the rule already has the menu-independent utility representation

$$\tilde{u}(x) = -p(x).$$

By Question 1.2 it must satisfy both α and β . We can also verify both directly.

For α , suppose $x \in C(A)$ and $x \in D \subseteq A$. Then

$$p(x) \leq p(z) \quad \text{for every } z \in A.$$

In particular this holds for every $z \in D$. Therefore x is still cheapest in D and

$$x \in C(D).$$

For β , suppose

$$x, y \in C(A), \quad A \subseteq B, \quad y \in C(B).$$

Because x and y are both cheapest in A ,

$$p(x) = p(y).$$

Because y is cheapest in B ,

$$p(y) \leq p(z) \quad \text{for every } z \in B.$$

Substituting $p(x) = p(y)$ gives

$$p(x) \leq p(z) \quad \text{for every } z \in B.$$

Hence x is also cheapest in B and

$$x \in C(B).$$

Thus both axioms hold even if cheapest-price ties make C multi-valued.

The distinction between the rationalizing utility $\tilde{u}$ and the DM's "true" utility is central here. The true utility is assumed to depend on price, energy efficiency, and attractiveness. Yet the observed choices are exactly as if the DM cared only about $-p$. Revealed preference from these choices therefore identifies a representation of the *heuristic*, not the underlying welfare function. In particular, the data cannot tell us how much the DM truly values efficiency or attractiveness because those attributes never affect choice.

5. The DM ranks the alternatives according to a utility function, and in any choice set chooses all median elements (note that if the median has more than one element all are chosen).

Solution. Assume for simplicity that utility ranks the alternatives strictly. Order any menu from lowest to highest utility. If the menu has an odd number of elements, the procedure chooses the unique middle-ranked element. If it has an even number, it chooses the two middle-ranked elements.

α **fails.** Take five alternatives satisfying

$$u(x_1) < u(x_2) < u(x_3) < u(x_4) < u(x_5).$$

On the five-element menu

$$A = \{x_1, x_2, x_3, x_4, x_5\},$$

the unique median is x_3 , so

$$C(A) = \{x_3\}.$$

Now contract to

$$D = \{x_3, x_4, x_5\}.$$

The alternative x_3 remains feasible, but it is now the lowest-ranked of the three alternatives.

The median of D is x_4 , so

$$C(D) = \{x_4\}.$$

Therefore

$$x_3 \in C(A), \quad x_3 \in D \subseteq A, \quad x_3 \notin C(D).$$

This violates α .

β **fails.** Use the four-element menu

$$A = \{x_1, x_2, x_3, x_4\}.$$

The two middle-ranked alternatives are x_2 and x_3 , so

$$C(A) = \{x_2, x_3\}.$$

Expand to

$$B = \{x_1, x_2, x_3, x_4, x_5\}.$$

The unique median is now x_3 , so

$$C(B) = \{x_3\}.$$

Thus

$$x_2, x_3 \in C(A), \quad A \subseteq B, \quad x_3 \in C(B), \quad x_2 \notin C(B).$$

This is exactly a violation of β .

Hence the median procedure satisfies neither α nor β . It therefore cannot be rationalized as maximizing one menu-independent utility function over the available alternatives. Notice that the utility ranking used to *define the median* is not a rationalizing utility for the observed choice rule: the DM deliberately chooses an interior rank rather than a utility maximum.

6. Rational Shortlisting: The DM is choosing between routes to drive home. For some routes they have live transit information and for some they do not. The DM makes decisions in two stages. First, they remove any routes such that there is a quicker available route according to the live transit information. Of the remaining routes they choose the shortest (in miles).

Solution. It helps to formalize the two criteria. Write

$$r \succ_T s$$

when the available live transit information establishes that route r is quicker than route s . Because live information is available only for some routes or comparisons, $\succ_T$ need not compare every pair. Write $d(r)$ for route r 's physical distance in miles.

For a menu A , the first-stage survivor set is

$$M(A) = \{r \in A : \text{there is no } s \in A \text{ with } s \succ_T r\}.$$

The second stage chooses the shortest route or routes in $M(A)$.

α **can fail.** Consider three routes x, y, z . Suppose the only live comparison relevant here is

$$y \succ_T z,$$

meaning live information establishes that y is quicker than z . Suppose x is not live-comparable with either of them. Let physical distances satisfy

$$d(z) < d(x) < d(y).$$

On the menu

$$A = \{x, y, z\},$$

route z is eliminated at stage 1 because $y \succ_T z$. Routes x and y survive. Since

$$d(x) < d(y),$$

stage 2 chooses

$$C(A) = \{x\}.$$

Now contract to

$$D = \{x, z\}.$$

The route y that previously eliminated z is gone. By construction there is no live comparison between x and z , so both survive stage 1. Since

$$d(z) < d(x),$$

stage 2 now chooses

$$C(D) = \{z\}.$$

Therefore

$$x \in C(A), \quad x \in D \subseteq A, \quad x \notin C(D).$$

Thus α fails.

The mechanism is worth making explicit. Contracting the menu can remove the route that was responsible for eliminating a third route. That third route can then re-enter the shortlist and defeat the formerly chosen route on miles.

β if the final rule is single-valued. If all physical distances relevant after shortlisting are distinct, or if a fixed tie-break rule is imposed, the procedure is a choice function. Then β holds automatically, exactly as in Question 2.3: $x, y \in C(A)$ implies $x = y$.

β if equal-distance ties are retained. If all shortest surviving routes are chosen, the procedure is a correspondence and β can fail. Consider routes x, y, z with the only relevant live comparison

$$z \succ_T x,$$

and no live comparison eliminating y . Let

$$d(x) = d(y) < d(z).$$

On the small menu

$$A = \{x, y\},$$

both routes survive stage 1 and have the same shortest distance, so

$$C(A) = \{x, y\}.$$

On the expanded menu

$$B = \{x, y, z\},$$

route x is eliminated because $z \succ_T x$. Routes y and z survive, and $d(y) < d(z)$, so

$$C(B) = \{y\}.$$

Hence

$$x, y \in C(A), \quad A \subseteq B, \quad y \in C(B), \quad x \notin C(B),$$

which violates β .

Therefore the robust conclusion is that rational shortlisting need not satisfy α , so it need not admit a standard menu-independent utility representation. If the rule is single-valued, β holds automatically; if distance ties are treated as multi-valued choice, β may fail as well.

Question 3 Let $\succeq$ be a complete relation on a non-empty set X , and for any $S \subset X$ finite define

$$C_{\succeq}(S) = \{x \in S \mid x \succeq y \text{ for all } y \in S\}.$$

1. Show that if $\succeq$ is transitive $C(S)$ is non-empty.

Solution. Because the definition of $C_{\succeq}(S)$ requires an alternative that weakly dominates *every* member of S , we need to show that completeness and transitivity guarantee such an alternative on every finite nonempty set.

Let

$$S = \{x_1, x_2, \dots, x_n\}.$$

We construct a “current best” alternative recursively.

Start with

$$m_1 = x_1.$$

For each $j = 2, \dots, n$, compare the current candidate m_{j-1} with the new alternative x_j .

If

$$m_{j-1} \succeq x_j,$$

set

$$m_j = m_{j-1}.$$

If instead

$$\text{not } (m_{j-1} \succeq x_j),$$

completeness implies

$$x_j \succeq m_{j-1},$$

and we set

$$m_j = x_j.$$

We prove by induction that after step j ,

$$m_j \succeq x_i \quad \text{for every } i \leq j.$$

For $j = 1$, this holds because completeness implies $x_1 \succeq x_1$.

Now suppose the claim holds for $j - 1$.

If $m_j = m_{j-1}$, then the induction hypothesis already gives

$$m_j = m_{j-1} \succeq x_i \quad (i < j),$$

and the condition under which we kept m_{j-1} gives

$$m_j = m_{j-1} \succeq x_j.$$

So the claim holds at step j .

If $m_j = x_j$, then completeness gave

$$x_j \succeq m_{j-1}.$$

By the induction hypothesis,

$$m_{j-1} \succeq x_i \quad (i < j).$$

Transitivity therefore implies

$$x_j \succeq x_i \quad (i < j).$$

Also $x_j \succeq x_j$. Hence again

$$m_j \succeq x_i \quad (i \leq j).$$

By induction, the final candidate m_n satisfies

$$m_n \succeq x_i \quad \text{for every } i = 1, \dots, n.$$

Since $m_n \in S$, this means

$$m_n \in C_{\succeq}(S).$$

Therefore

$$C_{\succeq}(S) \neq \emptyset.$$

The finiteness of S is what allows this sequential comparison procedure to terminate after finitely many steps.

2. We say a binary relation is acyclic if there is no k and set $x_1, \dots, x_k \in X$ such that

$$x_1 \succ x_2 \succ \cdots \succ x_k \succ x_1.$$

Show that acyclicity is a strictly weaker property than transitivity.

Solution. To show that acyclicity is *strictly weaker* than transitivity, we must establish two separate claims:

- (i) every transitive complete relation is acyclic;
- (ii) there exists an acyclic complete relation that is not transitive.

(i) Transitivity implies acyclicity. Suppose $\succeq$ is transitive. Recall that its strict part is defined by

$$x \succ y \iff x \succeq y \text{ and not } (y \succeq x).$$

Suppose, toward a contradiction, that there is a strict cycle

$$x_1 \succ x_2 \succ \cdots \succ x_k \succ x_1.$$

From the first $k - 1$ strict comparisons we obtain the weak comparisons

$$x_1 \succeq x_2, \quad x_2 \succeq x_3, \quad \dots, \quad x_{k-1} \succeq x_k.$$

Repeated transitivity implies

$$x_1 \succeq x_k.$$

But the final strict comparison

$$x_k \succ x_1$$

means, by definition,

$$x_k \succeq x_1 \quad \text{and} \quad \text{not } (x_1 \succeq x_k).$$

This contradicts the conclusion $x_1 \succeq x_k$. Hence no strict cycle can exist. Therefore transitivity implies acyclicity.

(ii) Acyclicity does not imply transitivity. Let

$$X = \{a, b, c\}.$$

Define a complete relation by

$$a \sim b, \quad b \sim c, \quad a \succ c.$$

Writing out what this means,

$$a \succeq b, \quad b \succeq a,$$

$$b \succeq c, \quad c \succeq b,$$

and

$$a \succeq c, \quad \text{but not } c \succeq a.$$

The relation is complete: every pair is comparable in at least one direction.

Its strict part contains only

$$a \succ c.$$

There is no strict edge leaving c that could return to a , and there are no strict comparisons involving b . Hence no strict cycle exists. The relation is acyclic.

However, transitivity fails. We have

$$c \succeq b \quad \text{and} \quad b \succeq a,$$

but

$$c \not\succeq a.$$

Thus the premise of transitivity holds for the chain $c \succeq b \succeq a$, while its conclusion fails.

Therefore acyclicity is weaker than transitivity, and because the converse implication fails, it is *strictly* weaker.

3. Show that if $\succeq$ is acyclic $C(S)$ is non-empty.

Solution. Here $\succeq$ is still assumed complete, but it need not be transitive. The key is that acyclicity of the strict part guarantees that a finite set contains an alternative that is not strictly dominated by any other alternative. Completeness will then convert “not strictly dominated” into the stronger condition required by $C_{\succeq}(S)$.

Let $\succ$ be the strict part of $\succeq$, and fix a finite nonempty set S .

Step 1: there is a $\succ$ -undominated element of S . Suppose not. Then every $x \in S$ is strictly dominated by some other member of S . Start from an arbitrary

$$x_1 \in S.$$

Because x_1 is supposedly dominated, choose $x_2 \in S$ such that

$$x_2 \succ x_1.$$

Because x_2 is also supposedly dominated, choose $x_3 \in S$ such that

$$x_3 \succ x_2.$$

Continue in this way, producing a sequence

$$x_1, x_2, x_3, \dots$$

with

$$x_{j+1} \succ x_j \quad \text{for every } j.$$

The sequence is infinite but S is finite. Therefore some alternative must repeat. That is, there exist indices $r < s$ such that

$$x_r = x_s.$$

The strict comparisons between the two occurrences give

$$x_s \succ x_{s-1} \succ \dots \succ x_{r+1} \succ x_r = x_s.$$

This is a strict cycle, contradicting acyclicity.

Therefore our supposition was false. There exists some $x^* \in S$ such that there is no $y \in S$ with

$$y \succ x^*.$$

Step 2: completeness implies $x^* \succeq y$ for every $y \in S$. Take an arbitrary $y \in S$. Suppose, toward a contradiction, that

$$x^* \not\succeq y.$$

Because $\succeq$ is complete, at least one of $x^* \succeq y$ or $y \succeq x^*$ must hold. Since the first one is assumed false, we must have

$$y \succeq x^*.$$

Together with $x^* \not\succeq y$, this is precisely

$$y \succ x^*.$$

But this contradicts the construction of x^* as an undominated element.

Hence the supposition was impossible, and

$$x^* \succeq y \quad \text{for every } y \in S.$$

Therefore

$$x^* \in C_{\succeq}(S),$$

so

$$C_{\succeq}(S) \neq \emptyset.$$

This proof also clarifies the role of completeness. Acyclicity gives an alternative with no strict dominator; completeness is what lets us conclude that this alternative weakly dominates every other element.

4. Show that if $C(S)$ is non-empty for every finite S , then $\succeq$ is acyclic.

Solution. The cleanest proof is by contraposition. Instead of directly proving

$$C_{\succeq}(S) \neq \emptyset \text{ for every finite } S \implies \succeq \text{ is acyclic,}$$

we prove the logically equivalent statement

$$\succeq \text{ is not acyclic} \implies C_{\succeq}(S) = \emptyset \text{ for some finite } S.$$

Suppose $\succeq$ is not acyclic. Then its strict part contains a cycle. Thus for some k there are alternatives

$$x_1, \dots, x_k$$

such that

$$x_1 \succ x_2 \succ \dots \succ x_k \succ x_1.$$

Let

$$S = \{x_1, \dots, x_k\}.$$

This set is finite.

We show that no element of S belongs to $C_{\succeq}(S)$. Consider any x_i .

For $i = 2, \dots, k$, its predecessor in the displayed cycle is x_{i-1} , and

$$x_{i-1} \succ x_i.$$

By definition of strict preference, this implies

$$\text{not } (x_i \succeq x_{i-1}).$$

Therefore x_i does not weakly dominate every element of S .

For x_1 , the final edge of the cycle gives

$$x_k \succ x_1,$$

so

$$\text{not } (x_1 \succeq x_k).$$

Thus x_1 also fails to weakly dominate every element of S .

Every member of S fails the defining condition for $C_{\succeq}(S)$. Hence

$$C_{\succeq}(S) = \emptyset.$$

This contradicts the hypothesis that $C_{\succeq}(S)$ is nonempty for every finite S . Therefore a strict cycle cannot exist, and $\succeq$ must be acyclic.

Question 4 One popular model of choice behavior in the marketing literature is choice with consideration sets. The basic idea is that, when presented with a choice set A , the decision maker only really thinks about a subset of those alternatives – $E(A)$ – and chooses the best of those.

Definition 1. We say a set of choice data can be explained as choice with consideration sets if there is (i) a utility function $u : X \rightarrow \mathbb{R}$ and (ii) a consideration set correspondence

$$E : 2^X \setminus \{\emptyset\} \rightarrow 2^X \setminus \{\emptyset\}$$

such that $E(A) \subseteq A$ and

$$C(A) = \arg \max_{x \in E(A)} u(x).$$

In other words, for each set A , $E(A)$ defines the set of alternatives that the decision maker considers. They then choose the best option from $E(A)$ according to u .

For simplicity, assume that we are dealing with choice functions (not correspondences), that there is no indifference, and that X is finite.

1. Show that, without further restrictions on E , a model of choice from consideration sets can explain **any** choice function.

Solution. Let C be an arbitrary choice function. Thus for every nonempty menu A ,

$$C(A) \in A.$$

We must construct a utility function u and a consideration correspondence E that reproduce exactly these choices.

Choose any utility function $u : X \rightarrow \mathbb{R}$ with no indifference. The ranking generated by this utility can be completely arbitrary; it will not matter for the construction.

For each nonempty menu A , define the consideration set to contain only the alternative that is actually chosen by C :

$$E(A) := \{C(A)\}.$$

This is a legitimate consideration set because $C(A) \in A$, so

$$E(A) \subseteq A,$$

and $E(A)$ is nonempty.

Now maximize u over $E(A)$. Since $E(A)$ contains only one alternative, that alternative is automatically the unique maximizer, whatever numerical value u assigns to it:

$$\arg \max_{x \in E(A)} u(x) = \arg \max_{x \in \{C(A)\}} u(x) = C(A).$$

This holds for every nonempty menu A . Hence the consideration-set model reproduces the original arbitrary choice function exactly.

The conclusion is that an unrestricted E imposes no testable restriction on single-valued choice data. Any apparent irregularity in choice can be explained by saying that, in each menu, the DM happened to consider only the alternative that was chosen. This is why restrictions on how $E(A)$ changes with A are economically important.

2. For the remainder of the question, add the restriction

$$\text{If } x \in B \subset A \text{ then } x \in E(A) \text{ implies } x \in E(B).$$

This means that consideration gets harder in larger choice sets, so if x is considered in the bigger set A then it should also be considered in the smaller set B .

Is this restriction implied by each of the following choice procedures?

- (a) In each choice set consider the three cheapest cars (or all the cars if there are fewer than three).

Solution. Yes, subject to the natural convention that ties in price are resolved by a fixed rule if necessary.

Suppose

$$x \in E(A), \quad x \in B \subset A.$$

Because $x \in E(A)$, x is one of the three cheapest cars in A (or A has fewer than three cars, in which case every car is considered).

If A has at least three cars, there can be at most two cars in A that are cheaper than x . Passing from A to the smaller menu B only removes cars. It cannot create any new car

cheaper than x . Therefore the number of cars in B that are cheaper than x is also at most two. Thus x remains among the three cheapest cars in B .

If B has fewer than three cars, then by the procedure every car in B is considered, so $x \in E(B)$ automatically.

In either case,

$$x \in E(A), \quad x \in B \subset A \implies x \in E(B).$$

Hence the restriction is satisfied.

- (b) In each choice set consider the fastest car and the car with the best safety rating.

Solution. Yes.

Suppose

$$x \in E(A), \quad x \in B \subset A.$$

By the procedure, x is considered in A for at least one of two reasons:

- (i) x is a fastest car in A ;
- (ii) x is a safest car in A .

If x is fastest in A , then no car in A is faster than x . Since every car in B is also in A , no car in B can be faster than x . Because $x \in B$, x remains a fastest car in B and is therefore considered.

If x is safest in A , the identical argument applies to the safety rating: removing other cars while retaining x cannot create a car with a higher safety rating than x . Hence x remains a safest car in B and is considered.

If x was both fastest and safest in A , either argument suffices. Thus

$$x \in E(A), \quad x \in B \subset A \implies x \in E(B).$$

The restriction is satisfied.

- (c) In each choice set consider cars that are unusual: You count up the number of cars in the choice set which are green, red, brown etc. and consider only those that are of the least common color.

Solution. No. Here the attention rule depends on *relative frequencies inside the current menu*, and those frequencies can change after contraction.

Consider a menu

$$A = \{g_1, g_2, r_1, r_2, r_3\},$$

where g_1, g_2 are green and r_1, r_2, r_3 are red. In A , green appears twice while red appears three times. Hence green is the least common color among the colors present. The procedure therefore considers the two green cars:

$$E(A) = \{g_1, g_2\}.$$

In particular,

$$g_1 \in E(A).$$

Now contract to

$$B = \{g_1, g_2, r_1\}.$$

The car g_1 remains feasible, so

$$g_1 \in B \subset A.$$

But in B , green appears twice while red appears once. Red is now the least common color, so

$$E(B) = \{r_1\}.$$

Therefore

$$g_1 \notin E(B).$$

We have found

$$g_1 \in E(A), \quad g_1 \in B \subset A, \quad g_1 \notin E(B),$$

which violates the restriction. Hence the unusual-color procedure does not imply the required monotonicity of consideration.

3. Is the following set of choices consistent with this model?

$$C(\{x, y, z\}) = x, \quad C(\{x, y\}) = y.$$

Solution. Yes. We need to show existence: construct one strict utility function u and one consideration correspondence E satisfying the restriction from part 2 and generating exactly these two observed choices.

Choose utilities satisfying

$$u(y) > u(x) > u(z).$$

For example, we could take

$$u(y) = 3, \quad u(x) = 2, \quad u(z) = 1.$$

Now define consideration sets by

$$E(\{x, y, z\}) = \{x, z\},$$

and for every proper nonempty subset $A \subsetneq \{x, y, z\}$ define

$$E(A) = A.$$

In particular,

$$E(\{x, y\}) = \{x, y\}.$$

Check the large menu. Since y is not considered there, the DM compares only x and z . Because

$$u(x) > u(z),$$

we obtain

$$C(\{x, y, z\}) = x.$$

Check the smaller menu $\{x, y\}$. Both alternatives are considered, and

$$u(y) > u(x),$$

so

$$C(\{x, y\}) = y.$$

Thus the choices are reproduced.

It remains to verify the consideration restriction. In the largest menu, the considered alternatives are x and z . Any smaller menu that contains x considers all of its elements, so it contains x in its consideration set. Likewise, any smaller menu that contains z considers z . The unconsidered alternative y in the large menu creates no restriction, because the implication in part 2 starts from $y \in E(A)$, which is false here. On every proper menu, $E(A) = A$, so every feasible alternative is considered and the restriction is immediate.

Hence this is a valid representation. The observed reversal

$$x \text{ chosen from } \{x, y, z\}, \quad y \text{ chosen from } \{x, y\}$$

is consistent with the restricted consideration-set model.

4. Show that the behavior in part 3 implies that $u(y) > u(x)$ if the model is correct.

Solution. The argument follows from two facts: a chosen alternative must be considered, and consideration is preserved when a menu contracts while retaining that alternative.

From

$$C(\{x, y, z\}) = x,$$

we know that x must belong to the consideration set of the large menu. Otherwise it could not be selected from that consideration set. Thus

$$x \in E(\{x, y, z\}).$$

Now consider the smaller menu

$$\{x, y\} \subset \{x, y, z\}.$$

It contains x . The restriction from part 2 therefore applies:

$$x \in E(\{x, y, z\}) \implies x \in E(\{x, y\}).$$

So x is definitely considered when the DM chooses from $\{x, y\}$.

But the observed choice on that smaller menu is

$$C(\{x, y\}) = y.$$

Therefore y is also considered, because a chosen alternative must belong to the consideration set. The DM thus chooses y while both x and y are in $E(\{x, y\})$.

The model says the DM chooses the utility maximizer from the consideration set. Hence

$$u(y) \geq u(x).$$

The problem additionally assumes no indifference. Therefore equality is impossible, and we obtain

$$u(y) > u(x).$$

This also explains how the large-menu choice of x can coexist with $u(y) > u(x)$: y must fail to be considered in the large menu. If y were considered there, it would beat x in utility and x could not be chosen.

5. Show that this model does not imply WARP.

Solution. To show that a model does *not* imply WARP, it is enough to provide one behavior that is generated by the model but violates WARP. Part 3 already gives exactly such behavior.

Let

$$A = \{x, y, z\}, \quad B = \{x, y\}.$$

Part 3 established that the restricted consideration-set model can generate

$$C(A) = x, \quad C(B) = y.$$

Both alternatives are feasible in both menus:

$$x, y \in A \cap B = \{x, y\}.$$

WARP says that if

$$x \in C(A) \quad \text{and} \quad y \in C(B),$$

with $x, y \in A \cap B$, then we must have

$$x \in C(B).$$

But $C(B) = y$ and choice is single-valued, so

$$x \notin C(B).$$

Therefore WARP is violated.

Because part 3 gave an explicit utility and consideration correspondence satisfying the model's restriction and producing these choices, this is a valid counterexample. Hence the consideration-set model does not imply WARP.

The source of the violation is attention, not an unstable utility ranking. Part 4 showed that $u(y) > u(x)$ is fixed. In the large menu, y can be ignored and x chosen; in the smaller menu, x remains considered but y enters consideration and is then chosen because it has higher utility.

6. Consider the following modification of the WARP* axiom of question 1.

Axiom 2. For any set S , there exists a b^* in S such that, for any T including b^* , if $C(T) \in S$ and $b^* = C(T')$ for some $T' \supset T$ then $C(T) = b^*$.

Show that the consideration set model described above does imply this behavior.

Solution. Fix an arbitrary nonempty set S . We need to find one alternative $b^* \in S$ that satisfies the axiom for every relevant menu T .

Because X is finite and the problem assumes no indifference, the utility function has a unique maximizer on S . Choose

$$b^* := \arg \max_{s \in S} u(s).$$

Thus

$$b^* \in S$$

and, for every $s \in S$ with $s \neq b^*$,

$$u(b^*) > u(s).$$

We prove that this b^* satisfies Axiom 2.

Take an arbitrary menu T such that

$$b^* \in T.$$

Suppose both additional premises in Axiom 2 hold:

$$C(T) \in S$$

and there exists a strict superset $T' \supset T$ such that

$$C(T') = b^*.$$

We must prove

$$C(T) = b^*.$$

Because b^* is chosen from T' , it must be considered there:

$$b^* \in E(T').$$

Also

$$b^* \in T \subset T'.$$

The consideration monotonicity restriction from part 2 therefore implies

$$b^* \in E(T).$$

This is the crucial step: if b^* was considered and chosen in the larger menu T' , then it cannot disappear from consideration when we contract to the smaller menu T while retaining it.

Now let

$$c := C(T).$$

The premise $C(T) \in S$ gives

$$c \in S.$$

Because c is chosen from T , the model says that c maximizes utility over $E(T)$. Since we have just shown that $b^* \in E(T)$,

$$u(c) \geq u(b^*).$$

On the other hand, b^* was defined to be the utility-maximizing element of S , and $c \in S$. Therefore

$$u(b^*) \geq u(c).$$

Combining the two inequalities gives

$$u(c) = u(b^*).$$

Because the problem assumes no indifference, two distinct alternatives cannot have the same utility. Hence

$$c = b^*.$$

Since $c = C(T)$,

$$C(T) = b^*.$$

This is exactly the conclusion required by Axiom 2.

Because S was arbitrary, the restricted consideration-set model implies Axiom 2 for every set S .

7. Consider the following variant of the above model. Rather than the condition in part 2, we have the condition

$$E(A \setminus \{x\}) = E(A) \text{ if } x \notin E(A).$$

In other words, if x is not considered at A , then removing it does not change the consideration set. Does the resulting model still satisfy the axiom from part 6?

Solution. No. The new condition is weaker in the direction that mattered for the proof of part 6. Part 6 used the fact that if b^* is considered in a larger set T' , then it must still be considered in a smaller set $T \subset T'$ that contains it. The new condition does not guarantee this. It only says that removing an *unconsidered* alternative leaves the consideration set unchanged.

We give a complete counterexample.

Let

$$X = \{x, y, a, b, z\}$$

and choose a strict utility ranking

$$u(x) > u(y) > u(a) > u(b) > u(z).$$

Thus there is no indifference, as required.

Define the consideration correspondence as follows. For every nonempty menu A except two special menus, let

$$E(A) = A.$$

For the two special menus, define

$$E(\{x, y, a\}) = \{y, a\},$$

and

$$E(\{x, y, b, z\}) = \{y, b, z\}.$$

We first check that this E satisfies the new condition.

For an ordinary menu with $E(A) = A$, there is no alternative $r \in A$ satisfying $r \notin E(A)$. Hence the condition has no antecedent to check and is automatically satisfied.

In the first special menu

$$A_1 = \{x, y, a\},$$

the only unconsidered alternative is x . Removing it gives

$$A_1 \setminus \{x\} = \{y, a\}.$$

This is an ordinary menu, so

$$E(\{y, a\}) = \{y, a\}.$$

Therefore

$$E(A_1 \setminus \{x\}) = \{y, a\} = E(A_1).$$

So the condition holds at A_1 .

In the second special menu

$$A_2 = \{x, y, b, z\},$$

the only unconsidered alternative is again x . Removing it gives

$$A_2 \setminus \{x\} = \{y, b, z\}.$$

This is ordinary, hence

$$E(\{y, b, z\}) = \{y, b, z\} = E(A_2).$$

So the condition also holds at A_2 .

Thus the variant attention restriction is satisfied on every menu.

Now take

$$S = \{x, y\}.$$

Axiom 2 would require that *at least one* of x or y can serve as the distinguished element b^* . We show that neither one works.

Candidate $b^* = x$ fails. Take

$$T = \{x, y, a\}$$

and

$$T' = \{x, y, a, z\}.$$

Clearly

$$x \in T \quad \text{and} \quad T' \supset T.$$

At the smaller menu T , the special consideration rule gives

$$E(T) = \{y, a\}.$$

Since

$$u(y) > u(a),$$

we obtain

$$C(T) = y \in S.$$

The larger menu T' is ordinary, so

$$E(T') = T'.$$

Because $u(x)$ is highest among all alternatives,

$$C(T') = x = b^*.$$

Therefore all premises of Axiom 2 are satisfied for candidate $b^* = x$:

$$x \in T, \quad C(T) = y \in S, \quad x = C(T') \text{ for } T' \supset T.$$

But the required conclusion fails because

$$C(T) = y \neq x.$$

Thus x cannot be the required b^* .

Candidate $b^* = y$ fails. Now take

$$T = \{x, y, b\}$$

and

$$T' = \{x, y, b, z\}.$$

Again

$$y \in T \quad \text{and} \quad T' \supset T.$$

The smaller menu T is ordinary, so all its alternatives are considered. Since

$$u(x) > u(y) > u(b),$$

we have

$$C(T) = x \in S.$$

The larger menu T' is the second special menu, so

$$E(T') = \{y, b, z\}.$$

Among those considered alternatives,

$$u(y) > u(b) > u(z),$$

so

$$C(T') = y = b^*.$$

Again all premises of Axiom 2 hold, but the conclusion fails:

$$C(T) = x \neq y.$$

Hence y cannot serve as b^* either.

Since

$$S = \{x, y\}$$

has no other candidate, there exists no $b^* \in S$ satisfying Axiom 2. Therefore the variant consideration-set model does *not* imply the axiom from part 6.

The contrast with part 6 is exact. The earlier restriction gave downward persistence of a *considered* alternative. The new restriction gives invariance only when an *unconsidered* alternative is removed. That is not enough to guarantee the key step $b^* \in E(T') \Rightarrow b^* \in E(T)$.